

Usman Karamat

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SUMMARY

Applied machine-learning engineer specializing in computer vision and edge deployment. Industry experience building and evaluating object-detection models on NVIDIA Jetson Orin (TensorRT) and training detection models on synthetic data to cut annotation cost. Background in mechanical engineering.

EXPERIENCE

Associate Software Engineer — Ransac Labs (Hazen.ai)

Sept 2025 – Present

Edge AI traffic-enforcement cameras (seatbelt & mobile-phone violation detection) · Lahore, Pakistan

- Develop the violation-detection module of an edge-deployed traffic-camera vision system, improving seatbelt and mobile-phone violation detection while reducing false positives.
- Benchmark and validate detection models on NVIDIA Jetson Orin using TensorRT engines; build evaluation pipelines computing precision, recall, and F1 across datasets, working with serialized model formats and Dockerized workflows.
- Drove a synthetic-data initiative to cut reliance on costly manual annotation: generated training imagery in Unity and trained PyTorch detection models, reaching near production-level seatbelt-detection performance from synthetic data alone.
- Experiment with model architectures and training strategies to balance detection accuracy against latency for real-time edge inference.

Software Quality Assurance Intern — CureMD

Jul 2025 – Aug 2025

CureMD · Lahore

- Completed ISTQB CTFL 4.0 certification; designed test cases using equivalence partitioning, boundary-value analysis, and flowcharting across the software development lifecycle.

Computer Vision Intern — National Center for Artificial Intelligence (NCAI)

Jun 2024 – Aug 2024

NCAI · Islamabad

- Built a YOLOv8 segmentation model to detect and classify corrosion severity (4 levels) on offshore oil-rig structures, supporting safer and lower-cost predictive maintenance.
- Trained on 8,000+ annotated images aggregated from open sources (Roboflow, OpenImages); achieving high segmentation accuracy with CUDA-accelerated inference to reduce processing time.

Freelance Math Expert (Solver & Reviewer) — Photomath, Inc.

Jun 2022 – Feb 2024

Remote · Math-learning platform (acquired by Google)

- Authored and reviewed step-by-step solutions for middle- and high-school mathematics on the Math Expert Marketplace, evaluating peer solutions for correctness and adherence to quality guidelines.
- Recognized as a top-performing Expert Solver (Apr 2023), earning priority task allocation for sustained solution quality over 20 months.

PROJECTS

Automated License-Plate Recognition

- Detection-plus-OCR pipeline (YOLOv8 + OCR) that automatically reads vehicle plates and logs entry/exit records for premises access control, replacing manual logging.
- Currently extending with stronger OCR, defined detection and end-to-end accuracy metrics, and an ONNX/TensorRT edge-deployment path.

EDUCATION

B.E. Mechanical Engineering — National University of Sciences & Technology (NUST)

Nov 2021 –

May 2025

Islamabad

- Completed CS minors in Object-Oriented Programming and Machine Learning alongside the core engineering degree.

TECHNICAL SKILLS

Focus: Computer vision, object detection, edge / embedded ML deployment, synthetic-data generation, model evaluation & benchmarking.

Tools & frameworks: Python, PyTorch, NumPy, Pandas, OpenCV, YOLOv8 / Ultralytics, TensorRT, ONNX, CUDA, Docker, Unity (synthetic data), NVIDIA Jetson Orin; C++, Java (familiar).

Certifications: Machine Learning Specialization – Stanford (Coursera); ISTQB CTFL 4.0.

Languages: English (professional), Urdu (fluent), Punjabi (native), Turkish (conversational), German (A1).